1 Circuit review

Phasors. $v(t) = V_{max}\cos(\omega t+\phi)$. Positive ϕ shifts the curve left and waveform occurs earlier. RMS value: $\tilde{V} = V_{rms} = V_{max}/\sqrt{2}$. $v(t) = \mathrm{Re}\{\tilde{V}e^{\omega t+\phi}\}$. Phasor drops $e^{\omega t}$: $V = \tilde{V}\angle\phi$. $j = \sqrt{-1} = 1\angle 90°$, and $1/j = -j$. Conjugate: $(r\angle\theta)^* = r\angle(-\theta)$.

Elements. Impedance $Z = R + jX$ [Ω], with resistance R and reactance X. Polar: $Z = |Z|\angle\theta$, $|Z| = \sqrt{R^2+X^2}$, $\tan\theta = X/R$. Admittance $Y = 1/Z = G + jB$ [S] Ohm's law: $V = IZ$, or $I = VY$. With $V = \tilde{V}\angle\phi_V$ and $I = \tilde{I}\angle\phi_I$, $\theta = \phi_V - \phi_I$. Inductor L: $v = L\,di/dt$, so $X_L = \omega L$. Capacitor C: $i = C\,dv/dt$, so $X_C = -1/(\omega C)$. Inductive ("lag"): $\theta > 0$, $\phi_I < \phi_V$, current occurs after voltage. Capacitive ("lead"): $\theta < 0$, $\phi_I > \phi_V$, current occurs before voltage.

Solving. Series: $Z_{eq} = Z_1 + Z_2 + \ldots$; same I. Parallel: $Y_{eq} = Y_1 + Y_2 + \ldots$, or $Z_{eq} = Z_1Z_2/(Z_1+Z_2)$ for two; same V. Voltage divider (series): $V_2 = V\cdot Z_2/(Z_1+Z_2)$. Current divider (parallel): $I_2 = I\cdot Z_1/(Z_1+Z_2)$. Norton: a current source $I_N = V_{Th}/Z_{Th}$ in parallel with $Z_N = Z_{Th}$. $v = V_{\mathrm{TH}} + R_{\mathrm{TH}}i$ with i coming into the circuit.

Transients. RC or RL circuit with a DC source: $x(t \geq 0) = x(\infty) + [x(0^+) - x(\infty)]\,e^{-t/\tau}$. Time constant $\tau = R_{Th}C$ (RC) or L/R_{Th} (RL), with R_{Th} seen by v_C or i_L. v_C and i_L cannot jump: $x(0^+) = x(0^-)$. $x(\infty)$: capacitor open, inductor short. Origin: particular solution (steady state) plus homogeneous $Ae^{-t/\tau}$, with A set by $x(0^+)$. Stored energy is $w(t) = \int_0^t p(t)\,dt = \int v(t)\,i(t)\,dt$. Capacitor: $i(t)\,dt = C\,dv$, so $w(t) = \int v(t)\,C\,dv = \frac{1}{2}Cv(t)^2$. Inductor: $v(t)\,dt = L\,di$.

2 Power

Definitions. Complex power $S = VI^* = P[W] + jQ[VAr]$ [VA] (¿0 absorbed). Polar: $S = \tilde{V}\tilde{I}\angle\theta$, where $\theta = \phi_V - \phi_I$. So $P = \frac{1}{2}V_{max}I_{max}\cos\theta = \tilde{V}\tilde{I}\cos\theta$, $Q = P\tan\theta$. Power factor $\mathrm{pf} = \cos\theta = P/|S|$. Lagging: $\theta > 0$, inductive, $Q > 0$, absorbs Q. Leading: $\theta < 0$, capacitive, $Q < 0$, supplies Q. Single element: $S = \tilde{I}^2Z = \tilde{V}^2Y^*$. $\theta = \pm\arccos(\mathrm{pf})$, positive if lagging.

3 Three-phase and per unit

Definitions. Balanced: phase b is phase a $\angle -120°$, phase c is phase a $\angle +120°$, for every voltage and current; solve phase a only. V_{ln}: phase to neutral, e.g. V_{an}. V_{ll}: phase to phase, $V_{ab} = V_{an} - V_{bn} = \sqrt{3}\,V_{an}\angle +30°$. Relationship holds for both connection types. Line current: in a conductor between source and load (I_a). Phase current: in one load element; equals the line current in Y (I_{ab} in Δ).

Connections. Y: element between phase and neutral; element voltage V_{ln}, element current $I_a = V_{an}/Z_Y$. Δ: element between two phases; element voltage V_{ll}, element current $I_{ab} = V_{ab}/Z_\Delta$. $I_a = I_{ab} - I_{ca} = \sqrt{3}\,I_{ab}\angle -30°$. Δ to Y: $Z_Y = Z_\Delta/3$ gives the same line currents.

Solving. Convert every Δ to Y and every V_{ll} to $V_{ln} = V_{ll}/\sqrt{3}\angle -30°$. Draw phase a: source V_{an}, series source and line impedances, Y-equivalent load, neutral return; shunt elements line to neutral. A load given as $|S_{3\phi}|$ (or P) and a pf at load voltage V_{ll}: $S_\phi = |S_{3\phi}|\angle\theta/3 = P/(3\mathrm{pf})\angle\theta$ with θ from pf; $Z_Y = \tilde{V}_{ln}^2/S_\phi^*$.

Power. $S_{3\phi} = 3V_{an}I_a^* = \sqrt{3}\,\tilde{V}_{ll}\tilde{I}_{line}\angle\theta$. Y: $|S|_{3\phi} = 3\tilde{V}_{ln}^2/|Z_Y|$. Δ: $|S|_{3\phi} = 3\tilde{V}_{ll}^2/|Z_\Delta| = 9\tilde{V}_{ln}^2/|Z_\Delta|$.

Per unit. S_b for the network, three-phase, typically 100 MVA; one V_b per zone, line-to-line, nominal. pu = actual/base; scales magnitues only. Single-phase $I_b = S_b/V_b$, $Z_b = V_b^2/S_b$. Three-phase $V_{b,ln} = V_{b,ll}/\sqrt{3}$, $I_b = S_b/(\sqrt{3}V_{b,ll})$, $Z_b = V_{b,ll}^2/S_b$; Base change: $Z_{pu,new} = Z_{pu,old}\,(V_{b,old}/V_{b,new})^2\,(S_{b,new}/S_{b,old})$. In pu, $|V_{ln}| = |V_{ll}|$ and $S_\phi = S_{3\phi}$. Across a transformer, set the other side's V_b by its line-to-line ratio; Z_{pu} is the same on both sides and the ideal transformer drops out.

4 Power flow and admittance matrices

Admittance matrix. Ground as reference; generators as current injections I_k; impedance loads and line charging as shunt admittances to ground; series impedances as $y_{km} = 1/Z_{km}$. KCL at each node gives $I_k = \sum_m Y_{km}V_m \implies I = YV$: Y_{kk} = sum of all admittances touching k; $Y_{km} = -y_{km}$,; symmetric, singular without a shunt. Branch current $y_{km}(V_k - V_m)$. $Y = A^\top\,\mathrm{diag}(y_{\mathrm{series}})A + \mathrm{diag}(y_{\mathrm{shunt}})$.

Power flow. $S_k = V_k\sum_m Y_{km}^*V_m^*$, generation minus load; with $V_k = \tilde{V}_k\angle\phi_k$, $Y_{km} = |Y_{km}|\angle\delta_{km}$: $P_k = \sum_m \tilde{V}_k\tilde{V}_m|Y_{km}|\cos(\phi_k - \phi_m - \delta_{km})$, Q_k the same with sin. Bus types: PQ (load), PV (generator), slack (one; $\tilde{V}$, $\phi = 0$ given). Implicit (Newton): unknowns $x = [\phi$ (PQ, PV); $\tilde{V}$ (PQ)], mismatch $f(x) = [P_k(x) - P_k^{given}$ (PQ, PV); $Q_k(x) - Q_k^{given}$ (PQ)], $x \leftarrow x - J^{-1}f(x)$ from $\tilde{V} = 1$, $\phi = 0$, with $J = [\partial P/\partial\phi,\ \partial P/\partial\tilde{V};\ \partial Q/\partial\phi,\ \partial Q/\partial\tilde{V}]$. Explicit (after): slack P, Q and PV-bus Q from their own equations. $P_{loss} = \sum_k P_k = \sum_{\mathrm{lines}}|I_{km}|^2R_{km}$.

5 Power line models

Models. Length ℓ: $Z = (r + jx)\ell$, $Y = jb\ell$. Short line: Z only. π model: Z in series, $Y/2$ to neutral at each end.

Setup. Sending end S, receiving end R. Load as parallel G and jB: $Y_L = S_\phi^*/\tilde{V}_R^2 = (P_\phi - jQ_\phi)/\tilde{V}_R^2$, i.e. $R_L = \tilde{V}_R^2/P_\phi$, $X_L = \tilde{V}_R^2/Q_\phi$; $Y_R = Y_L + Y_{sh}/2$ (π model), $Y_R = Y_L$ (short line). $I_{line} = V_RY_R$, $V_S = V_R(1 + ZY_R)$, $I_S = I_{line} + V_SY_{sh}/2$. $P_{loss} = 3\,\mathrm{Re}(V_SI_S^*) - P_{load} = 3|I_{line}|^2R$. $VR = (|V_{R,NL}| - |V_{R,FL}|)/|V_{R,FL}|$ at fixed V_S; no load is $Y_L = 0$: short line $V_{R,NL} = V_S$, π model $V_{R,NL} = V_S/(1 + ZY/2)$.

6 Magnetism and magnetic circuits

Magnetism. Magnetic field intensity H [A/m]. Permeability $\mu = \mu_r\mu_0$ [H/m]; $\mu_0 = 4\pi\times 10^{-7}$ H/m (air). Magnetic flux density $B = \mu H$ [T = Wb/m^2]. Magnetic flux $\Phi = BA$ [Wb] through a cross-section of area A [m^2]. Flux linkage $\lambda = N\Phi = Li$ [Wb·turn] for a coil of N turns. Faraday: $v = d\lambda/dt = N\,d\Phi/dt$ [V]. Ampère: $\oint H\,dl = I_{enc}$, the current passing through the

loop formed by the path. For the mean core path of length l [m] with uniform H, and a coil of N turns carrying i [A] threading the loop: $Hl = Ni$. Ni is the magnetomotive force (MMF) F [A·turn].

Magnetic circuit. Electric $\iff$ Magnetic: EMF V [V] $\iff$ MMF $F = Ni$ [A·turn]; current I [A] $\iff$ flux Φ [Wb]; resistance R [Ω] $\iff$ reluctance $\mathcal{R} = l/(\mu A)$ [A/Wb], for a segment of length l, area A, permeability μ; KVL $\iff$ $\sum F - \sum \mathcal{R}\Phi = 0$ around a loop (Ampère); KCL $\iff$ $\sum \Phi = 0$ at a junction (flux continuity); Ohm's law $V = IR$ $\iff$ $F = \Phi\mathcal{R}$; series and parallel combine like resistance. Coil = MMF source of value Ni, direction by the right-hand rule. One $\mathcal{R}$ per segment of distinct l, μ, or A. Air gap: $\mathcal{R}_{gap} = l_{gap}/(\mu_0 A)$ dominates, since $\mu_0 \ll \mu_{core}$.

Inductance. Coil of N turns on a single core of length l, area A: $L = \lambda/i = (N\Phi)/i = N^2/\mathcal{R}$ [H]. Several segments: $L = N^2/\mathcal{R}_{total}$, with $\mathcal{R}_{total}$ seen by the coil.

Saturation. B cannot exceed B_{sat} anywhere in the core. Procedure: $F = Ni_{peak}$; Φ from the magnetic circuit; $B_k = \Phi_k/A_k$ in each segment k; $B_{sat} \geq \max_k B_k$, in the smallest A_k carrying the largest Φ_k.

Losses. Core, eddy: flux induces voltage in the core, voltage drives current, current heats; $\propto f^2B^2$; fix with laminations and silicon steel. Core, hysteresis: area of the B–H loop per cycle; $\propto f\times$ area; fix by material. Winding, skin: AC current crowds to the surface; fix with Litz wire. Winding, proximity: one conductor's field induces eddy currents in its neighbors, worse with more layers; fix with fewer or interleaved layers, Litz wire.

7 Transformers

Two windings on one core, N_1 turns on the primary and N_2 on the secondary. Faraday on the primary, $V_1 = j\omega N_1\Phi$, fixes the core flux Φ. The same Φ links the secondary, so $V_2 = j\omega N_2\Phi$ and $V_1/N_1 = V_2/N_2$, with or without a load. A load draws I_2; its MMF N_2I_2 would change Φ, so the primary draws the I_1 that cancels it. Dot: current into the dotted terminal of either winding drives flux the same way around the core. Define I_1 into the primary dot and I_2 out of the secondary dot; then V_1, V_2 are both + at the dots and MMF balance is $N_1I_1 - N_2I_2 = \mathcal{R}\Phi$. Autotransformer: one winding with a tap; primary and secondary share turns. Tap changer: moves the tap to change N_1/N_2 for voltage control.

Ideal. $\mu \to \infty \implies \mathcal{R} \to 0 \implies N_1I_1 = N_2I_2$. Turns ratio $a = N_1/N_2$, so $V_1 = aV_2$, $I_1 = I_2/a$, and $S_1 = V_1I_1^* = S_2$.

Equivalent circuit (nonideal). Refer across the ideal a:1: $Z_{eq1} = a^2Z_{eq2}$. Circuit: primary terminals $\to Z_1 = R_1 + jX_1 \to (R_c \parallel jX_m) \to$ ideal a:1 $\to Z_2 = R_2 + jX_2 \to$ secondary terminals; $X_m = \omega L_m$ with $L_m = N_1^2/\mathcal{R}_{tot}$ (energy stored in core), R_c core loss, R_k winding resistance, X_k leakage reactance. $Z_{eq1} = Z_1 + a^2Z_2$; $Z_{sh} = R_c \parallel jX_m$. $I_1 = V_1/\big(Z_1 + Z_{sh} \parallel (a^2Z_2 + Z_L')\big)$; with $Z_1 = Z_2 = 0$, $I_1 = V_1/Z_L' + V_1/Z_{sh}$.

Solving. Efficiency $\eta = P_{\text{out}}/(P_{\text{out}} + P_{\text{core}} + P_{\text{copper}})$. $P_{\text{core}} = P_{oc}$, constant at rated voltage. $P_{\text{copper}} = \tilde{I}_1^2R_1 + \tilde{I}_2^2R_2 = \tilde{I}_2^2R_{eq2}$. Voltage regulation VR: $V_{2,NL} = V_1/a$ and $V_{2,FL} = V_1/a - I_2Z_{eq2}$ at the given load; $VR = 0$ for an ideal source. Leading pf: the component of I_2Z_{eq2} along $V_{2,FL}$ can be negative, so VR can be negative.

Three-phase. Three single-phase transformers, each side Y or Δ; bank rating $S_{3\phi} = 3S_{\text{transformer}}$. Winding quantities: Y side $V_w = V_{ln}$, $I_w = I_{\text{line}}$; Δ side $V_w = V_{ll}$, $I_w = I_{\text{line}}/\sqrt{3}$. Per-transformer turns ratio $a = V_{1,w}/V_{2,w}$; line-to-line ratio $V_{1,ll}/V_{2,ll} = a$ (Y–Y, Δ–Δ), $a/\sqrt{3}$ (Δ–Y), $\sqrt{3}a$ (Y–Δ). Δ–Y and Y–Δ shift voltages and currents by 30° between sides; magnitudes and powers unaffected. Procedure: convert line to winding quantities and $S_\phi = S_{3\phi}/3$; solve the single-phase problem on one transformer; convert back; add the 30° only if a phasor is needed relative to the other side.

8 Machines

Rotor: rotating part. Stator: fixed part. Field winding: makes the main flux from DC current i_f or a permanent magnet. Armature winding: has the EMF induced in it; its current i_a carries the converted power. $E_{\text{electric}} = E_{\text{mech}} + E_{\text{field}} + E_{\text{loss}}$. Four facts: current in a wire makes a field (transformer primary); a wire carrying i in field B feels $\vec{F} = i\,\vec{l}\times\vec{B}$ (motor); a changing field makes a voltage in a coil (transformer secondary); a wire moving at $\vec{v}$ in field B has EMF $e = (\vec{v}\times\vec{B})\cdot\vec{l}$ (generator). Left-hand rule, motor: pointer B, middle i, thumb F.

DC machines. Stator: field, fixed flux from a permanent magnet or DC field winding. Rotor: armature, fed DC through brushes; the commutator reverses each coil's current to keep one diretion of torque. (+) cheap drive, stable. (−) commutator wear.

Circuit: terminals $\to R_a \to L_a \to$ back EMF $E = k_e\omega$, so $V = R_ai_a + L_a\,di_a/dt + k_e\omega$. Steady state drops L_a term. Motor ($i_a \to R_a \to E$): $V = R_ai_a + k_e\omega$. Generator ($E \to R_a \to i_a$): $V = k_e\omega - R_ai_a$. Back EMF constant $k_e = (V - i_aR_a)/\omega$ [V·s/rad], from a no-load test. Torque constant k_t [N·m/A] in $T = k_ti_a$. Power balance $P_{\text{mech}} = Ei_a = T\omega \implies k_e = k_t$ and $i_a = T/k_t$. $P_{\text{in}} = Vi_a$. Torque–speed: $\omega = V/k_e - TR_a/(k_ek_t)$, a line from no-load speed $\omega_0 = V/k_e$ ($T = 0$) to stall torque $T_0 = k_tV/R_a$ ($\omega = 0$).

Synchronous machines. Stator: armature, three-phase winding; its currents make a field rotating at ω_s and carry the AC power. Rotor: field winding with DC i_f through slip rings (or a permanent magnet); its flux locks to the rotating field, so the shaft turns at ω_s. p pole pairs: $\omega_s = \omega_e/p = 2\pi f_e/p$ [rad/s], since the stator field advances one pole pair, $1/p$ of a turn, per electrical cycle. At standstill the torque alternates at f_e and averages zero (solution: squirrel cage or ramp f_e) (+) pf set by i_f; speed fixed by f_e. (−) not self-starting; needs a DC field supply. Per-phase circuit: EMF $E_a \to R_s \to jX_s \to$ terminal V_a; generator has I_a from machine to grid so $(E_a - V_a) = (R_s + jX_s)I_a$; reverse direction for motor. Typically set $R_s = 0$. $E_a = k_fi_f\omega_e\angle\delta = (L_f/\sqrt{2})\,i_f\omega_e\angle\delta$; $\delta = \phi_{E_a} - \phi_{V_a}$ for a generator; reverse for motor so δ always > 0. $S_{3\phi} = 3V_aI_a^* = 3V_a((E_a - V_a)/jX_s)^*$ with $\tilde{V}_a\angle 0$ and $\tilde{E}_a\angle\delta$. Torque–speed: vertical line at ω_s. Torque–δ: $T = P_{3\phi}/\omega_s = 3\tilde{V}_a\tilde{E}_a\sin\delta/(X_s\omega_s)$, a sine, max at $\delta = 90°$, unstable past it. $T = (\pi/2)\,p\,\Phi_rF_f\sin\delta$ with air-gap flux per pole Φ_r and field MMF $F_f = N_fi_f$; raise T by Φ_r (limited by B_{sat}) or F_f (limited by field heating).